Since $\frac{|G|}{\dim V} \in \mathbb{Q}$ and $\sum_{C_i} \lambda_i \overline{\chi_V(g_{C_i})} \in \overline{\mathbb{Z}}$, by Proposition 5.2.5, $\frac{|G|}{\dim V} \in \mathbb{Z}$. $\square$

Exercise 5.3.3. Strengthen the result of Exercise 5.1.7: show that all nontrivial irreducible representations of a group of odd order are of complex type. (Use that any representation of quaternionic type is even-dimensional).

5.4. Burnside's theorem

Definition 5.4.1. A group G is called **solvable** if there exists a series of nested normal subgroups

$$\{e\} = G_1 \triangleleft G_2 \triangleleft \ldots \triangleleft G_n = G$$

where G_{i+1}/G_i is abelian for all $1 \leq i \leq n-1$.

Remark 5.4.2. Such groups are called solvable because they first arose as Galois groups of polynomial equations which are solvable in radicals.

Theorem 5.4.3 (Burnside)**.** *Any group G of order $p^a q^b$, where p and q are primes and $a, b \geq 0$, is solvable.*

This famous result in group theory was proved by the British mathematician William Burnside in the early 20th century, using representation theory (see Section 5.5 and [**Cu**]). Here is this proof, presented in modern language.

Before proving Burnside's theorem, we will prove several other results which are of independent interest.

Theorem 5.4.4. *Let V be an irreducible representation of a finite group G and let C be a conjugacy class of G with* $\gcd(|C|, \dim(V)) = 1$. *Then for any $g \in C$, either $\chi_V(g) = 0$ or g acts as a scalar on V.*

The proof will be based on the following lemma.

Lemma 5.4.5. *If $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n$ are roots of unity such that $\frac{1}{n}(\varepsilon_1 + \varepsilon_2 + \cdots + \varepsilon_n)$ is an algebraic integer, then either $\varepsilon_1 = \cdots = \varepsilon_n$ or $\varepsilon_1 + \cdots + \varepsilon_n = 0$.*